SET

Yves Zenou Junjie Zhou

Paper version: February 17, 2026

Equilibrium uniqueness and representation

In nonlinear models, standard sufficient conditions for uniqueness can be difficult to verify.

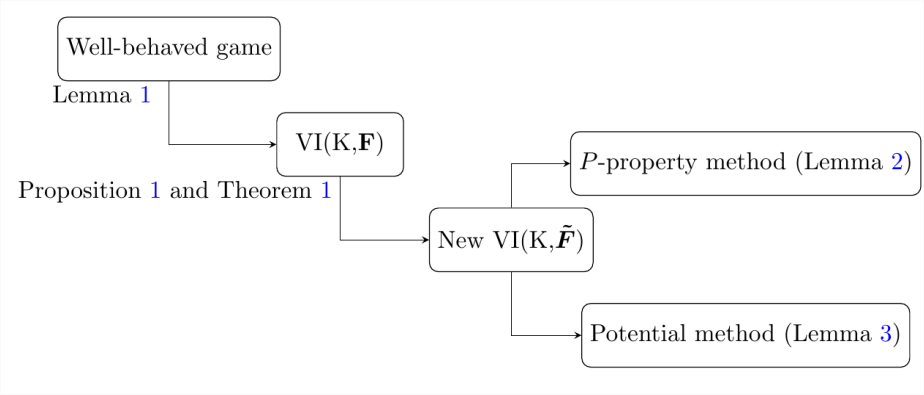
Some of this difficulty may come from how the equilibrium conditions are written.

The central question

How can we transform nonlinear equilibrium to simplify analysis, without changing the equilibrium set?

The paper develops sign-equivalent transformations (SET) to reveal useful mathematical structure in equilibrium problems.

The paper's map



An equilibrium is a best-response fixed point. The figure follows its VI representation through a sign-equivalent transformation to two methods for studying uniqueness.

Example: different nonlinear best responses

Two players choose $x_i \geq 0$, with $K = \mathbb{R}^2_+$. Write $a(x_1) = e^{2-x_1} > 0$.

Payoffs

$$u_1(x) = x_1(1 + x_2^2) - \tfrac{1}{2}x_1^2,$$
$$u_2(x) = 3a(x_1)^2 x_2 - a(x_1)x_2^2 - \tfrac{1}{3}x_2^3.$$

Own-action curvature is $\partial_{11} u_1 = -1$ and $\partial_{22} u_2 = -2(a + x_2) < 0$.

Best responses

$$B_1(x_2) = 1 + x_2^2, \qquad B_2(x_1) = e^{2-x_1}.$$

Player 1 responds positively; player 2 responds negatively. Both responses are positive.

Example: the original VI

The negative marginal payoffs define $F_i = -\partial_i u_i$.

$$F(x) = \begin{pmatrix} x_1 - 1 - x_2^2 \\ x_2^2 + 2ax_2 - 3a^2 \end{pmatrix}.$$

$$x^* \in \mathrm{NE} \quad \Longleftrightarrow \quad \langle y - x^*, F(x^*) \rangle \geq 0 \quad \text{for every } y \in K.$$

One possible uniqueness test asks whether $DF(x)$ is a P-matrix throughout K.

The two-dimensional P-matrix test

A matrix J is a P-matrix if every principal minor is positive:

$$J_{11} > 0, \qquad J_{22} > 0, \qquad \det J > 0.$$

Example: the original P-matrix test fails

The original Jacobian is

$$DF(x) = \begin{pmatrix} 1 & -2x_2 \\ 2a(3a - x_2) & 2(a + x_2) \end{pmatrix}.$$

Its diagonal entries are positive, but its determinant need not be:

$$\det DF(x) = 2(a + x_2) + 4ax_2(3a - x_2).$$

At the interior feasible point $\widehat{x} = (2 + \ln 2, 3)$, we have $a = 1/2$ and

$$DF(\widehat{x}) = \begin{pmatrix} 1 & -6 \\ -3/2 & 7 \end{pmatrix}, \qquad \boxed{\det DF(\widehat{x}) = -2 < 0.}$$

The original Jacobian fails the global P-matrix test. This failure alone says nothing about the number of equilibria.

Example: best responses are not a contraction

The joint best-response map and its Jacobian are

$$B(x) = \begin{pmatrix} 1 + x_2^2 \\ e^{2-x_1} \end{pmatrix}, \qquad DB(x) = \begin{pmatrix} 0 & 2x_2 \\ -e^{2-x_1} & 0 \end{pmatrix}.$$

At the interior equilibrium $x^* = (2, 1)$,

$$DB(x^*) = \begin{pmatrix} 0 & 2 \\ -1 & 0 \end{pmatrix}, \qquad \lambda = \pm i\sqrt{2}, \qquad \rho(DB(x^*)) = \sqrt{2} > 1.$$

Any contraction with modulus $q < 1$ would require, in its induced norm,

$$\rho(DB(x^*)) \leq \|DB(x^*)\| \leq q < 1,$$

a contradiction. Thus B is not a contraction in any fixed norm, even locally.

Example: a nonlinear sign-equivalent transformation

Complete the square in the second component:

$$F_2 = x_2^2 + 2ax_2 - 3a^2 = (x_2 + a)^2 - (2a)^2.$$

Use the strictly increasing function $h(t) = \sqrt{t}$ on $(0, \infty)$:

$$\widetilde{F}_2(x) = \sqrt{(x_2 + a)^2} - \sqrt{(2a)^2} = x_2 - a.$$

Keeping $\widetilde{F}_1 = F_1$ gives

$$\widetilde{F}(x) = \begin{pmatrix} x_1 - 1 - x_2^2 \\ x_2 - e^{2-x_1} \end{pmatrix}, \qquad \operatorname{sgn} \widetilde{F}_i(x) = \operatorname{sgn} F_i(x) \quad \text{on } K.$$

Both square-root inputs are strictly positive on K. Rectangularity gives

$$\mathrm{NE} = \mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).$$

Example: the transformed P-matrix test

The transformed Jacobian is

$$D\widetilde{F}(x) = \begin{pmatrix} 1 & -2x_2 \\ e^{2-x_1} & 1 \end{pmatrix}.$$

For every $x \in K = \mathbb{R}^2_+$, its three principal minors satisfy

$$(D\widetilde{F})_{11} = (D\widetilde{F})_{22} = 1 > 0,$$

$$\det D\widetilde{F}(x) = 1 + 2x_2 e^{2-x_1} \geq 1 > 0.$$

Thus $D\widetilde{F}$ is a P-matrix throughout the same feasible domain. The transformed VI has at most one solution, so the game has at most one equilibrium.

This is the P-matrix route in the paper's map; no potential is needed.

Example: existence and the unique equilibrium

Existence is explicit: at $x^* = (2, 1)$,

$$\widetilde{F}(2,1) = \begin{pmatrix} 2 - 1 - 1^2 \\ 1 - e^{2-2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Every best response is strictly positive, so no equilibrium lies on the boundary.

$$\boxed{\mathrm{NE} = \mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}) = \{(2,1)\}.}$$

- Original *DF*: the global *P*-matrix test fails.
- Best responses: no contraction in any fixed norm.
- After SET: a global *P*-matrix certificate gives uniqueness.

Individual optimality and the VI

Definition 1 (Well-behaved game)

A normal-form game $\Gamma = (u_i, K_i)_{i \in N}$ is well-behaved if, for every player i:

1. $K_i \subseteq \mathbb{R}^{m_i}$ is closed and convex.
2. u_i is twice continuously differentiable (C^2).
3. $u_i(\cdot, x_{-i})$ is concave on K_i for every feasible x_{-i}.

Lemma 1 (Nash equilibrium and the VI)

For a well-behaved game, set $K = \prod_i K_i$ and $F_i = -\nabla_{x_i} u_i$. Then, for $x^* \in K$,

$$x^* \in \mathrm{NE} \quad \Longleftrightarrow \quad \langle y - x^*, F(x^*) \rangle \geq 0 \quad \text{for every } y \in K.$$

The VI as familiar boundary conditions

Lemma 2

Let $K = \prod_i [a_i, b_i]$, with $-\infty \leq a_i < b_i \leq +\infty$. For $x^* \in K$,

$$\langle y - x^*, F(x^*) \rangle \geq 0 \quad \text{for every } y \in K$$

$$\Updownarrow$$

$$\left.\begin{aligned} x_i^* = a_i &\implies F_i(x^*) \geq 0 \quad \text{(lower bound)}, \\ a_i < x_i^* < b_i &\implies F_i(x^*) = 0 \quad \text{(interior)}, \\ x_i^* = b_i &\implies F_i(x^*) \leq 0 \quad \text{(upper bound)}. \end{aligned}\right\} \quad \text{for every } i.$$

With $F_i = -\partial_i u_i$, these are the familiar constrained first-order conditions.

Only the sign of each component matters, including at the boundary.

Sign equivalence on rectangular domains

Definition 2 (Sign equivalence)

F and $\widetilde{F}$ are sign-equivalent on K if $\operatorname{sgn} F_i(x) = \operatorname{sgn} \widetilde{F}_i(x)$ for every i and every $x \in K$.

Theorem 1 (Part (ii): rectangular domains)

If $K = \prod_i [a_i, b_i]$ and $F, \widetilde{F}$ are sign-equivalent on K, then

$$\operatorname{Sol}(K, F) = \operatorname{Sol}(K, \widetilde{F}).$$

Two constructions of a sign-equivalent operator

Proposition 1 (SET 1 and SET 2)

Finite compositions of the following componentwise operations preserve signs.

1. **Positive rescaling:**

$$\widetilde{F}_i(x) = \alpha_i(x) F_i(x), \qquad \alpha_i(x) > 0 \quad \text{on all of } K.$$

2. **An increasing transformation:** if $F_i = A_i - B_i$ and h_i is strictly increasing,

$$\widetilde{F}_i = h_i(A_i) - h_i(B_i) \quad \Longrightarrow \quad \operatorname{sgn} \widetilde{F}_i = \operatorname{sgn} F_i.$$

For strictly decreasing h_i, reverse the order: $h_i(B_i) - h_i(A_i)$.

The rescaling factor may depend on the full profile, as in oligopoly pricing. Our worked example uses $h(t) = \sqrt{t}$ on strictly positive arguments. Logs and inverses require valid domains.

The role of rectangularity: a counterexample

Consider a VI with a shared constraint rather than separate action intervals:

$$K = \{x \geq 0 : x_1 + x_2 \leq 1\}, \qquad F = (-1, -1), \qquad \widetilde{F} = (-2, -1).$$

Both operators have the same component signs at every feasible point. For a constant operator, the VI minimizes its linear objective over K.

$$\mathrm{Sol}(K, F) = \{x \geq 0 : x_1 + x_2 = 1\}, \qquad \mathrm{Sol}(K, \widetilde{F}) = \{(1, 0)\}.$$

Discussion

Why does changing the relative weight of the two coordinates matter when they compete for a shared resource?

This is a coupled-constraint VI illustration. Its domain is not a product of intervals.

The P-condition and constrained uniqueness

A useful global P-condition is: for every $x \neq y$ in K,

$$\max_i (x_i - y_i)\big[F_i(x) - F_i(y)\big] > 0.$$

Suppose x and y both solve the VI on a rectangle. Feasible one-coordinate deviations give, for every i,

$$(x_i - y_i)F_i(x) \leq 0, \qquad (x_i - y_i)F_i(y) \geq 0.$$

Subtracting implies

$$(x_i - y_i)\big[F_i(x) - F_i(y)\big] \leq 0 \quad \text{for every } i.$$

This contradicts the P-condition unless $x = y$. The argument covers constrained solutions as well as interior zeros.

What a better representation can reveal

Let K be convex and $\widetilde{F}$ continuously differentiable.

Lemma 2 (Part (ii): P-property)

On a rectangle K, if $D\widetilde{F}(x)$ is a P-matrix for every $x \in K$, then the VI has at most one solution.

A P-matrix has positive principal minors.

Used for: oligopoly pricing, peer effects, CES complements.

Lemma 3 (Potential)

If $D\widetilde{F}$ is symmetric and positive semidefinite on K, the VI minimizes a convex potential with $\nabla\Phi = \widetilde{F}$. If it is positive definite, there is at most one solution.

Used for: network games, public goods, multiplex interactions.

Existence needs its own argument in each application.

Selected Applications

Each application follows the same comparison.

1. Write the original equilibrium operator F and its Jacobian.
2. Apply a sign-equivalent transformation.
3. Inspect the new Jacobian and state the uniqueness condition.

- **Oligopoly pricing:** positive rescaling and a P-matrix.
- **CES network complements:** logarithms and a P-matrix.
- **Network public goods:** an inverse function and a symmetric Jacobian.

Oligopoly: the original equilibrium operator

Firm i chooses $p_i \in [a_i, b_i]$, with $c_i \leq a_i < b_i$, and earns

$$\pi_i(p) = (p_i - c_i)Q_i(p), \qquad Q_i > 0, \quad Q_{i,i} := \frac{\partial Q_i}{\partial p_i} < 0.$$

The outside-option price p_0 is fixed. Let $K = \prod_i [a_i, b_i]$. The original operator is negative marginal profit:

$$\boxed{F_i(p) = -Q_i(p) - (p_i - c_i)Q_{i,i}(p).}$$

Under E1, log-profit is strictly concave for $p_i > c_i$. At $p_i = c_i$, marginal profit is $Q_i > 0$. Hence the marginal-profit signs characterize best responses, and

$$\mathrm{NE} = \mathrm{Sol}(K, F).$$

The demand restrictions

Write $\ell_i = \log Q_i$. Demand is positive, C^2, and decreasing in own price.

Assumption E1 (Own-price log-concavity)

$$\frac{\partial^2 \ell_i}{\partial p_i^2} \leq 0.$$

Assumption E2 (Cross-price log-supermodularity)

$$\frac{\partial^2 \ell_i}{\partial p_i \partial p_j} \geq 0 \quad (j \neq i, \text{ including the outside option } j = 0).$$

Assumption E3 (Common-price-shift invariance)

$$Q_i(p_1 + t, \ldots, p_n + t, p_0 + t) = Q_i(p_1, \ldots, p_n, p_0), \quad i = 0, \ldots, n.$$

Oligopoly: the original Jacobian

Write $Q_{i,j} = \partial Q_i / \partial p_j$ and $Q_{i,ij} = \partial^2 Q_i / \partial p_i \partial p_j$. Differentiating $F_i = -Q_i - (p_i - c_i) Q_{i,i}$ gives

$$\boxed{\begin{aligned} (DF)_{ii} &= -2Q_{i,i} - (p_i - c_i) Q_{i,ii}, \\ (DF)_{ij} &= -Q_{i,j} - (p_i - c_i) Q_{i,ij} \qquad (j \neq i). \end{aligned}}$$

The own-price demand slope contributes a positive term to the diagonal. The second-derivative term has no such guaranteed sign.

Even the diagonal entries need not be positive. The original Jacobian therefore need not pass the global P-matrix test.

Oligopoly: SET removes the demand-slope scale

Divide each component by the positive number $-Q_{i,i}$:

$$\widetilde{F}_i(p) = \frac{F_i(p)}{-Q_{i,i}(p)} = p_i - c_i - \mu_i(p), \qquad \mu_i(p) = \frac{Q_i}{-Q_{i,i}} = -\frac{1}{\partial_{p_i} \log Q_i}.$$

Thus $\widetilde{F}_i$ is actual markup minus demand-implied markup. The multiplier $1/(-Q_{i,i})$ is positive throughout the feasible domain, so

$$\operatorname{sgn} \widetilde{F}_i = \operatorname{sgn} F_i.$$

Since the price domain is rectangular,

$$\boxed{\mathrm{NE} = \mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).}$$

This includes equilibrium prices at a floor or cap.

Oligopoly: the transformed Jacobian

Let $J = D\widetilde{F} = I - D\mu$. For every price coordinate, including p_0,

$$\frac{\partial \mu_i}{\partial p_j} = \frac{\partial^2 \ell_i / \partial p_i \partial p_j}{(\partial \ell_i / \partial p_i)^2}.$$

E1 and E2 therefore give

$$\boxed{J_{ii} = 1 - \partial_{p_i} \mu_i \geq 1, \qquad J_{ij} = -\partial_{p_j} \mu_i \leq 0 \quad (j \neq i).}$$

Here J contains only the n strategic price coordinates.

The transformed Jacobian has a positive diagonal and nonpositive cross entries. E3 controls their combined magnitude.

Oligopoly: diagonal dominance and uniqueness

E3 implies $\sum_{j=0}^{n} \partial_{p_j} \mu_i = 0$. Consequently,

$$\boxed{J_{ii} - \sum_{j \neq i} |J_{ij}| = 1 + \partial_{p_0} \mu_i \geq 1 > 0.}$$

Positive diagonal entries and strict row diagonal dominance make J a P-matrix at every feasible price vector.

Proposition 4 (Oligopoly pricing)

Under E1–E3, the pricing game on $K = \prod_i [a_i, b_i]$, with $a_i \geq c_i$, has at most one pure-strategy Nash equilibrium.

Lemma 2(ii) gives uniqueness, including binding price bounds.

CES complements: the original operator and Jacobian

On a connected, nonempty undirected network without self-loops, let

$$u_i = a_i x_i - \tfrac{1}{2}x_i^2 + \delta x_i S_i(x), \qquad S_i(x) = \left(\sum_j g_{ij} x_j^{\beta}\right)^{1/\beta}, \quad a_i, \delta, \beta > 0.$$

Every equilibrium is positive and satisfies $x_i = T_i(x) := a_i + \delta S_i(x)$. Thus the original operator and Jacobian on $\mathbb{R}^n_{++}$ are

$$F_i(x) = x_i - a_i - \delta S_i(x), \qquad \boxed{DF(x) = I - \delta DS(x),}$$

$$(DS(x))_{ij} = g_{ij} x_j^{\beta-1} S_i(x)^{1-\beta}.$$

The aggregate's derivatives depend on the levels and relative sizes of efforts. For $\beta < 1$, we do not assert differentiability at zero.

CES complements: SET and log coordinates

First apply the increasing logarithm to the two positive terms:

$$\widetilde{F}_i(x) = \log x_i - \log T_i(x), \qquad \operatorname{sgn} \widetilde{F}_i(x) = \operatorname{sgn}[x_i - T_i(x)] .$$

Then make the one-to-one change of variables $z_i = \log x_i$:

$$H_i(z) = \widetilde{F}_i(e^z) = z_i - \log T_i(e^z), \qquad z \in \mathbb{R}^n.$$

Every equilibrium has $x_i \geq a_i > 0$, so

$$\boxed{x^* \in \mathrm{NE} \quad \Longleftrightarrow \quad H(\log x^*) = 0.}$$

The logarithmic comparison is SET. The change of coordinates is a separate step.

CES complements: the new Jacobian and uniqueness

The Jacobian in log coordinates is

$$\boxed{DH(z) = I - W(z),} \qquad W_{ij}(z) = \frac{\delta S_i(e^z)}{a_i + \delta S_i(e^z)} \frac{g_{ij} e^{\beta z_j}}{\sum_k g_{ik} e^{\beta z_k}}.$$

Since there are no self-loops, $W_{ii} = 0$. Also $W_{ij} \geq 0$ and

$$\sum_j W_{ij}(z) = \frac{\delta S_i(e^z)}{a_i + \delta S_i(e^z)} < 1.$$

Hence, for every $z \in \mathbb{R}^n$,

$$(DH)_{ii} - \sum_{j \neq i} |(DH)_{ij}| = 1 - \sum_j W_{ij} > 0.$$

The positive diagonal and strict row dominance give a P-matrix. By Lemma 2(ii), H has at most one zero, so the game has at most one equilibrium. This is the conditional-uniqueness argument used in Theorem 4.

Public goods: the original operator and Jacobian

On an undirected network without self-loops, let $s = (I + \delta G)x$ and $u_i(x) = b(s_i) - c(x_i)$, with $x_i \geq 0$ and $0 < \delta \leq 1$.

Assumption 2 (Benefits and costs)

$b' > 0$, $b'' < 0$, $c'(u) > 0$ for $u > 0$, and $c'' \geq 0$, with $b'(0) > c'(0)$ and $\lim_{u\to\infty} b'(u) < \lim_{u\to\infty} c'(u)$.

Negative marginal payoffs give

$$F_i(x) = c'(x_i) - b'(s_i), \qquad \boxed{DF(x) = C(x) + A(x)(I + \delta G),}$$

where

$$C(x) = \operatorname{diag}(c''(x_i)), \qquad A(x) = \operatorname{diag}(-b''(s_i)) \succ 0.$$

For $j \neq i$, $(DF)_{ij} = -\delta g_{ij} b''(s_i)$. Its row-dependent benefit curvature generally makes DF asymmetric.

Public goods: SET through an inverse function

Work on an equilibrium-preserving rectangle $K \subset [0, k^*]^n$ where $c'(K_i) \subset \text{range}(b')$, with $b'(k^*) = c'(k^*)$. The inverse is finite on this domain. Define

$$h(u) = (b')^{-1}(c'(u)).$$

Because b' is strictly decreasing,

$$F_i(x) = b'(h(x_i)) - b'(s_i) \quad \sim_{\text{sign}} \quad s_i - h(x_i).$$

The transformed operator is therefore

$$\boxed{\widetilde{F}(x) = (I + \delta G)x - h(x).}$$

Rectangularity preserves the constrained VI solutions:

$$\text{NE} = \text{Sol}(K, F) = \text{Sol}(K, \widetilde{F}).$$

Public goods: the transformed Jacobian

Differentiate $b'(h(u)) = c'(u)$:

$$h'(u) = \frac{c''(u)}{b''(h(u))} \leq 0.$$

The new Jacobian is

$$\boxed{D\widetilde{F}(x) = I + \delta G - \mathrm{diag}(h'(x_i)).}$$

Its diagonal and cross entries are

$$(D\widetilde{F})_{ii} = 1 - h'(x_i) \geq 1, \qquad (D\widetilde{F})_{ij} = \delta g_{ij} = (D\widetilde{F})_{ji} \quad (j \neq i).$$

SET removes the row-dependent benefit curvature from the cross effects. The Jacobian is now symmetric, making the potential route available.

Public goods: positive definiteness and uniqueness

Since $G = G^\top$ and $-h'(x_i) \geq 0$, for every $x \in K$,

$$D\widetilde{F}(x) \succeq [1 + \delta\lambda_{\min}(G)]I.$$

Thus $1 + \delta\lambda_{\min}(G) > 0$ makes the transformed Jacobian positive definite. Symmetry gives a potential, and positive definiteness makes it strictly convex.

Theorem 5 (Public-good equilibrium)

Under Assumption 2, if $1 + \delta\lambda_{\min}(G) > 0$, the game has a unique Nash equilibrium.

The Jacobian calculation supplies the uniqueness argument. The theorem's existence argument is separate.

Jacobian comparison across applications

Application	Before SET	After SET
Oligopoly	First and second demand derivatives	$I - D\mu$ Strict row dominance gives a P-matrix.
CES complements	$I - \delta DS$	$I - W$, with $W \geq 0$ and row sums below one. A P-matrix in log coordinates.
Public goods	$C + A(I + \delta G)$	$I + \delta G - \text{diag}(h')$ Symmetric and positive definite under the spectral condition.

Main conclusions

A general method

On rectangular domains, sign-equivalent transformations preserve the entire VI solution set and can simplify the Jacobian used to study uniqueness.

- Oligopoly and CES complements use a transformed P-matrix.
- Public goods use symmetry and positive definiteness.
- The worked example has a unique equilibrium even though its original global P-matrix test fails and its best responses are not a contraction.

The useful representation depends on the model. Existence remains a separate question.

Backup: a coordinate proof of Theorem 1(ii)

Let $K = \prod_i K_i$ and suppose $x^* \in \mathrm{Sol}(K, F)$. For every i and $t \in K_i$,

$$(t - x_i^*) F_i(x^*) \geq 0.$$

Sign equivalence implies the same inequality for $\widetilde{F}_i(x^*)$:

$$\begin{cases} F_i(x^*) = 0 & \Longrightarrow \ \widetilde{F}_i(x^*) = 0, \\ F_i(x^*) > 0 & \Longrightarrow \ t - x_i^* \geq 0 \text{ and } \widetilde{F}_i(x^*) > 0, \\ F_i(x^*) < 0 & \Longrightarrow \ t - x_i^* \leq 0 \text{ and } \widetilde{F}_i(x^*) < 0. \end{cases}$$

Consequently, for every $x \in K$,

$$\langle x - x^*, \widetilde{F}(x^*) \rangle = \sum_i (x_i - x_i^*) \widetilde{F}_i(x^*) \geq 0.$$

Thus $\mathrm{Sol}(K, F) \subseteq \mathrm{Sol}(K, \widetilde{F})$. Interchanging the two operators gives the reverse inclusion.